Which bits can we drop?

Component-wise quantization of a learned image codec

PULSE-MSE | 14.18M parameters | 8 quality settings | one weight file

We keep 39 times more quality than uniform INT8 at the same size.

21.677 MB
60.4 percent smaller. Loss is 0.94 percent.

18.664 MB
65.9 percent smaller. Loss is 1.60 percent.

39x
better than uniform INT8 at the same size

One precision for all parts is the wrong tool

the parts are not equal

Group	Parameters	Share	What it means
Encoder E0-E5	12.13M	85.6%	four blocks alone hold 11.6 MB
Entropy H0-H3	1.34M	9.4%	takes 73 percent of a decode
Decoder D0-D8	708k	5.0%	small, but one 4.7k block loses 8 percent

Sensitivity does not follow size.

A 2.89M encoder block loses 0.5 percent at 8 bits.

A 4.7k renderer block loses 8 percent at the same precision.

One weight file serves all eight settings. A choice per setting is not deployable.

So the task is an allocation problem: one precision per part, decided once.

artifacts/component_inventory.json

How we measured it

69,233 measurements. One part at a time.

7 formats

FP16, BF16, FP8-E4M3, FP8-E5M2, INT8, INT4, FP4-E2M1

19 parts x 8 settings

weights only. All other parts stay in FP32.

Kodak and CLIC

24 and 41 images. All eight quality settings.

Penalty = 100 x [(1 + dbpp/100) x 2^(-dPSNR/slope) - 1]

One number holds a quality change and a bitrate change. Zero means equal to FP32.

Additive: 0.96-1.05

Single-part costs add up, so the solver can sum them.

FP32 gate passes

The reference bitstream matches at all eight settings.

scripts/pipeline.py (metric) | analyze_additivity.py (check)

Three parts refuse 8 bits

13 of 19 are safe. The three that are not are small.



Intolerant at INT8

H3 entropy step: 38.15%

H0 hyper encoder: 10.09%

D5 renderer: 8.06%

Together 3.8 percent of parameters.

Cheap at INT8

E2 / E3: 0.51 / 0.43%

E1 / E4: 1.35 / 0.71%

D1: 0.12%

results/analysis/component_precision_penalty_heatmap.png

We solved the choice as an integer program

one variable per part. Shared across all eight settings.

Why shared variables

One weight file serves all eight settings. A different assignment per setting would need eight files.

Constraints

One quality limit for each setting. Eight limits in total.

Objective

Minimise bytes. Not latency: we have no measured latency table per format, so an objective using one would optimize invented data.

Point	Size	Kodak	CLIC
1% champion	**21.677 MB**	**0.67%**	**0.85%**
1.8% variant	**18.664 MB**	**1.27%**	**1.60%**
INT8 tier + QAT	14.401 MB	1.85%	2.87%

The knee is between 1.6 and 1.8 percent: 2.5 MB falls away in one step.

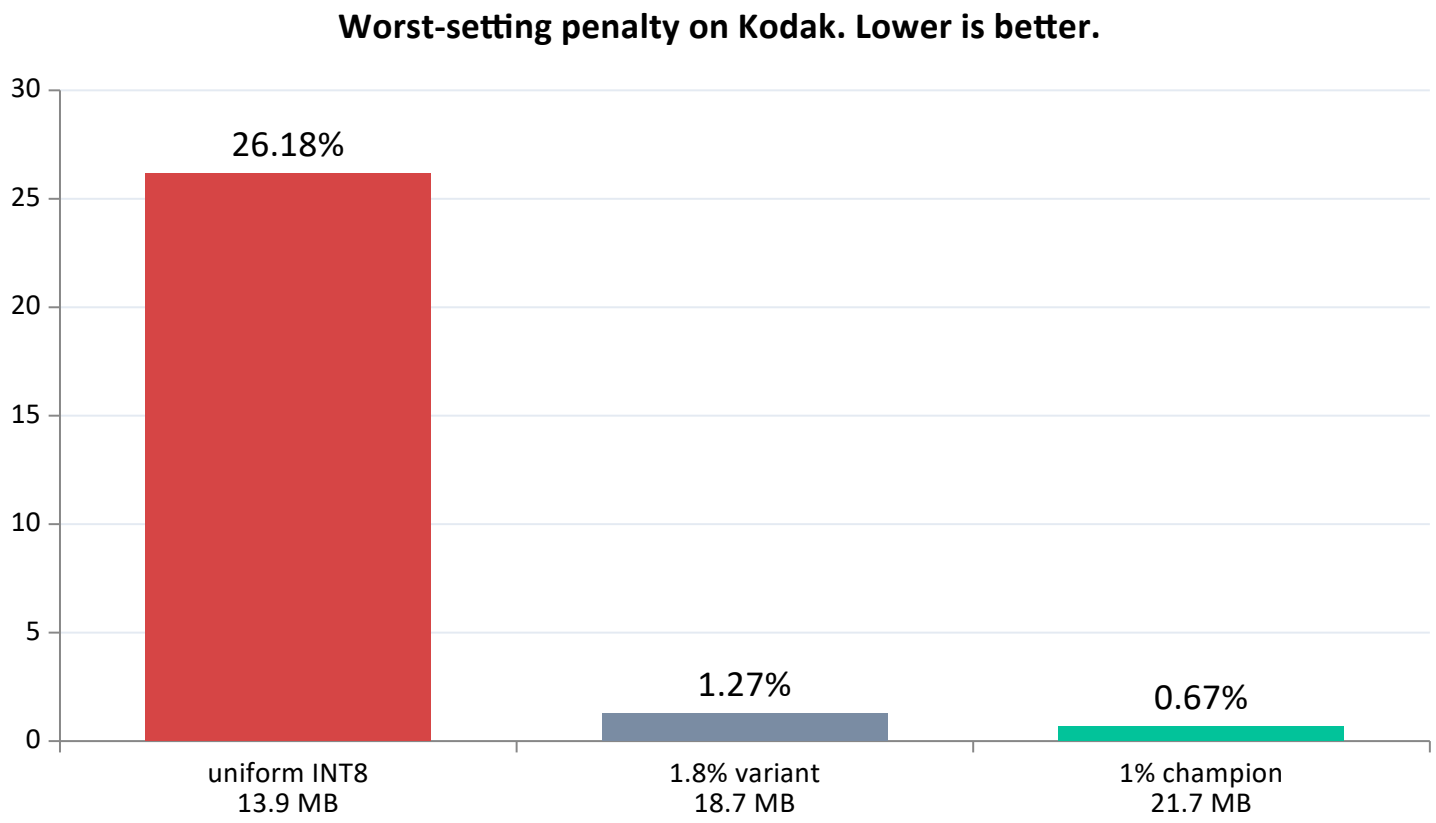
The 1.8 percent assignment beats the 2 percent one we first chose. It is 0.275 MB larger and better on all three sets.

Solved with HiGHS. A greedy baseline matches it within a few percent.

results/final/SUBMISSION_POINTS.md | results/night/ilp_fine_sweep.json

The comparison that decides it

uniform INT8 is smaller and 39 times worse



26.18%

uniform INT8 at 13.9 MB

0.67%

our champion at 21.7 MB

Uniform INT8 also varies 2.9 times across datasets. The mixed model varies 1.4 times. A limit on the total gives quality and stability.

results/mixed/ | 7 configurations, 3 datasets

4-bit failed. That is why we fine-tuned.

Group scaling helped by 5 to 22 times. Not enough.

Part	INT8 cost/MB	INT4 cost/MB
E3	0.14	31.4
E2	0.17	26.1
E4	0.24	31.2
E1	0.46	157.5
D5	1712	not viable

Cost per MB is the decision number: penalty points paid for one megabyte saved.

Group-32 scaling cut INT4 weight error by 40 percent. The cost stays about 100 times that of the 8-bit step.

So 4-bit is not worth shipping alone. Can fine-tuning pay for it?

Constrained, not penalised

Hold bitrate at its old value. Minimise distortion. A fixed penalty drifts in both directions.

Entropy stays frozen

The bundle holds integer CDF tables. Move the weights and encoder and decoder disagree.

12,000 steps

2.2 percent of the pretraining steps. A repair, not a new training.

results/qat/WHY_THESE_COMPONENTS.md | scripts/qat_run.py

QAT repairs what the choice leaves

61 to 98 percent. And zero where nothing was left.

Tier	Size	Before	2k	12k	Removed
INT8 tier	14.401 MB	8.61%	8.61%	3.36%	61.0%
INT4, 2 blocks	13.572 MB	119.18%	25.27%	20.53%	82.8%
INT4, 4 blocks	11.369 MB	1771.71%	84.22%	41.84%	97.6%
ILP 2% assignment	18.389 MB	1.936%	not run	1.936%	0.0%

0.0%

gain on the assignment the solver optimized

61-98%

of damage removed on hand-chosen tiers

Fine-tuning replaces a good allocation. It does not extend it.

At 2,000 steps the INT8 tier gained nothing. We first concluded that QAT only helps severe damage. At 12,000 steps it removed 61 percent. The first result was a step-budget effect.

This is our most useful finding, not a failure: the repair is as large as the damage that the choice left behind.

results/final/qat_attempt/summary.csv | results/night/qat_int8_12k/summary.csv

We did not get a speedup. Here is why.

Quantization accelerates the part that is not the bottleneck.

A decode takes 88 ms.

Entropy coding takes 80 ms. The network takes 8 ms.

Convolutions are 9 percent of a decode and 27 percent of a round trip.

Quantization cannot fix the part that dominates.

+4.79%

latency for the mixed model. 95 percent CI +1.27 to +9.80.

-0.39%

energy change. The interval includes zero.

We make no speed claim for FP8, INT4 or FP4. These formats run as simulated quantization.

Only FP16 and BF16 have real kernels on this GPU. They save energy, not time.

A real integer speedup needs integer kernels. That is an engineering task, not a policy choice.

results/metrics/ | paired interleaved benchmark, 50 pairs per cell

What we would ship, and what would change it

21.677 MB

champion. 60.4 percent smaller, 0.94 percent loss.

18.664 MB

size-first variant. 65.9 percent smaller, 1.60 percent loss.

Do not use uniform INT8 at 13.9 MB. It is smaller and 39 times worse.

Do not use 4-bit without fine-tuning.

Limits we state first

Checkpoint selection under-reports the QAT gain. The probe said no change; the full test said 8.61 to 3.36 percent.

Weights only. No activation quantization, so W8A8 is not tested.

4-bit is simulated, so its speed is not measured.

One entropy lane. The largest unused latency lever.

QAT uses one seed and 100 images. The direction holds. The exact numbers are single-run.

Granularity is the lever. Fine-tuning repairs what a choice leaves. An optimal choice leaves nothing.